

AGENTIC MENTORING FOLLOW-UP AND PARENT COMMUNICATION ASSISTANT

1. INTRODUCTION

1.1 Background And Problem Context

Educational institutions continuously strive to improve student academic performance and overall development. Mentors play a crucial role in identifying students who require academic guidance and emotional support. However, tracking attendance records, internal assessment marks, previous academic performance, and maintaining communication with parents manually can be challenging and time-consuming.

Traditional mentoring approaches often rely on spreadsheets and manual documentation, which may lead to delayed interventions and inconsistent follow-ups. As the number of students increases, mentors find it difficult to prioritize students requiring immediate attention.

Agentic Artificial Intelligence provides an opportunity to automate and enhance the mentoring process. By enabling intelligent agents to analyze student records, make decisions, remember previous interactions, and generate personalized recommendations, institutions can improve mentoring efficiency and student outcomes.

The proposed "Agentic Mentoring Follow-up and Parent Communication Assistant" utilizes Agentic AI principles to assist mentors in identifying at-risk students, generating professional parent communication messages, maintaining interaction history, and planning follow-up actions.

1.2 Motivation

The motivation behind selecting this project stems from the increasing workload faced by faculty mentors in monitoring students' academic progress. During regular mentoring sessions, mentors often spend significant time analyzing attendance records and assessment scores while maintaining parent communication logs.

This project combines the concepts of Agentic AI with a real-world educational application. It demonstrates how intelligent agents can autonomously analyze information, provide recommendations, and support decision-making processes.

Furthermore, the project aligns with the growing adoption of AI-driven educational technologies aimed at improving student engagement and academic success.

1.3 Objectives

The primary objectives of the proposed system are:

- To design an Agentic AI system for academic mentoring.
- To analyze student attendance and internal assessment performance.
- To identify students requiring immediate mentoring attention.

- To generate professional parent communication messages.
- To maintain mentoring interaction history.
- To recommend suitable mentor interventions.
- To create follow-up plans and review schedules.
- To demonstrate the practical application of Agentic AI concepts.

1.4 Scope

The proposed system focuses on academic mentoring activities within educational institutions.

The system covers:

- Student attendance analysis.
- Internal assessment evaluation.
- Risk categorization of students.
- Parent communication generation.
- Follow-up recommendation generation.
- Storage of interaction history.

The system does not cover:

- Automatic attendance collection.
- Integration with institutional ERP systems.
- Financial or administrative activities.
- Psychological counseling diagnosis.
- Automated parent calling facilities.

2. LITERATURE REVIEW

2.1 Existing Systems

Existing System 1: Traditional Manual Mentoring

Most institutions use manual mentoring practices involving spreadsheets, paper records, and periodic meetings. Mentors manually analyze attendance and academic records before contacting parents.

Limitations:

- Time-consuming.

- Difficult to scale.
- Lack of automated recommendations.
- Inconsistent follow-up mechanisms.

Existing System 2: Learning Management Systems (LMS)

Platforms such as Moodle and Google Classroom provide academic tracking features and assignment monitoring.

Advantages:

- Centralized data management.
- Easy access to academic records.

Limitations:

- Limited mentoring intelligence.
- No autonomous decision-making.
- Lack of personalized parent communication.

Existing System 3: Student Early Warning Systems

Several institutions implement early warning systems to identify academically vulnerable students using predictive models.

Advantages:

- Early risk detection.
- Data-driven interventions.

Limitations:

- Focus primarily on prediction.
- Limited communication capabilities.
- Minimal interaction memory.

2.2 Key Concepts

Agentic AI

Agentic AI refers to intelligent systems capable of perceiving information, reasoning, making decisions, and performing actions autonomously to achieve defined objectives. Unlike conventional AI systems, Agentic AI exhibits proactive behavior and utilizes memory to improve future interactions.

Risk Assessment

Risk assessment involves identifying students who may require academic intervention based on predefined criteria such as attendance percentage, internal assessment scores, and backlog information.

Parent Communication

Effective parent communication strengthens collaboration between educational institutions and families. Timely communication ensures that parents remain informed about academic concerns and participate in corrective actions.

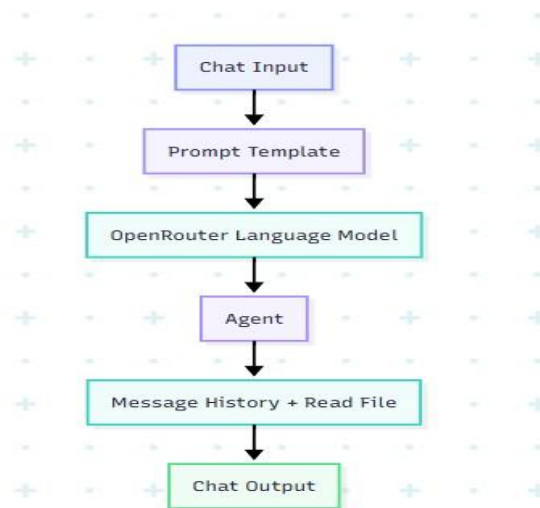
Conversational Memory

Memory mechanisms allow AI agents to retain previous interactions and contextual information. This capability enables continuity in mentoring sessions and supports personalized follow-up recommendations.

3.SYSTEM / PROJECT DESIGN

3.1 System Architecture Diagram

Architecture Flow:



3.2 Architecture Explanation

Chat Input

The Chat Input component accepts student details such as attendance, IA marks, backlog information, and mentor observations.

Prompt Template

The Prompt Template provides structured instructions guiding the agent to analyze student records and generate mentoring outputs.

OpenRouter Language Model

The language model performs reasoning tasks and supports intelligent response generation.

Agent

The Agent acts as the core decision-making component. It analyzes academic information, assesses risks, and determines appropriate actions.

Message History

This component stores previous mentoring interactions and enables retrieval of contextual information.

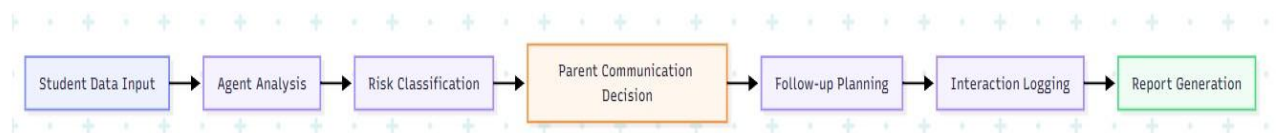
Read File Tool

The Read File tool allows mentors to upload and analyze student information stored in external documents.

Chat Output

The final mentoring report, recommendations, and communication messages are displayed through Chat Output.

3.3 Flowchart / Use Case Explanation Flow:



The flow illustrates how the agent processes academic information, generates recommendations, and maintains continuity through stored interactions.

4. IMPLEMENTATION DETAILS

4.1 Technologies Used

- Langflow
- Agentic AI Concepts
- OpenRouter
- Large Language Models
- Prompt Engineering
- Message History

4.3 Step-by-Step Implementation

Module 1: Input

The mentor provides student information through the Chat Input interface or uploads a file containing academic records.

Module 2: Processing

The Prompt Template structures the task instructions. The language model and agent analyze the data, classify risks, and determine suitable interventions.

Module 3: Output

The system generates:

- Risk Level
- Parent Communication Decision
- Parent Message
- Mentor Recommendation

4.4 Prompts Used

Prompt:

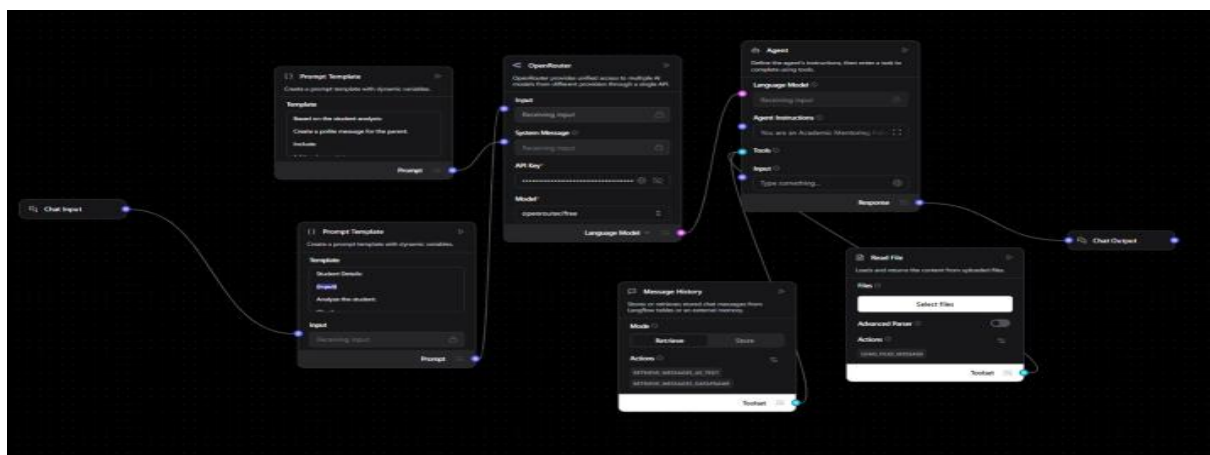
"You are an Academic Mentoring Follow-up and Parent Communication Assistant.

Analyze student records and generate:

1. Risk Level
2. Parent Contact Requirement
3. Parent Communication Message
4. Follow-up Recommendation

5. RESULTS AND ANALYSIS

5.1 Output Screenshots



5.2 Result Explanation

The developed system successfully analyzed student academic records and generated structured mentoring reports. The agent accurately identified students requiring intervention based on attendance and academic performance indicators.

The generated parent communication messages maintained professionalism while clearly conveying concerns and suggested actions.

The follow-up recommendations enabled mentors to establish timely review schedules and maintain continuity in mentoring activities.

5.3 Advantages

- Reduces manual mentoring workload.
- Enables early identification of at-risk students.
- Improves consistency in parent communication.
- Supports personalized follow-up recommendations.

5.4 Limitations

- Dependent on the quality of input data.
- Limited to predefined mentoring scenarios.
- Does not integrate directly with institutional databases.

5.5 Future Enhancements

- ERP integration.
- Automated email and SMS notifications.
- Mobile application support.
- Predictive analytics for academic forecasting.

7. CONCLUSION

The Agentic Mentoring Follow-up and Parent Communication Assistant demonstrates the practical application of Agentic AI principles within the educational domain. The project successfully automates several mentoring activities including student performance analysis, parent communication generation, and follow-up planning.

Through the integration of intelligent agents, memory mechanisms, and prompt engineering, the proposed system enhances mentoring efficiency and promotes timely interventions. The project also provided valuable insights into the capabilities and limitations of Agentic AI systems.

Overall, this assignment strengthened the understanding of autonomous agents and their potential to solve real-world problems in educational environments.

8. REFERENCES

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- [4] Langflow Documentation and User Guide.
- [5] Brownlee, J., Machine Learning Mastery with Python.
- [6] IEEE Research Papers on AI in Education.
- [7] Research Articles on Early Warning Systems for Student Performance Prediction.

APPENDIX

Appendix A: Complete Langflow Workflow Screenshots.

Appendix B: Prompt Templates Used.

Appendix C: Sample Student Input Records.

Appendix D: Sample Parent Communication Messages.

Appendix E: Follow-up Recommendation Outputs.

Appendix F: Exported Langflow Configuration File.

Appendix G: Additional Experimental Results.